ChatGPT

What is the clinical significance of BRAF V600E in Melanoma with Trametinib in CIViC?

Understood. I will look up the clinical significance of the BRAF V600E mutation in melanoma in relation to trametinib using CIViC's evidence and interpretation data, ensuring I capture relevant recent updates. I'll gather details and report back with findings.

Worked for 6 minutes >

Clinical-interpretation databases such as CIViC base variant annotations on published trials and regulatory approvals. In cutaneous melanoma the BRAF-V600E point mutation constitutively activates the MAP-kinase pathway and is found in roughly half of melanomas pmc.ncbi.nlm.nih.gov. Because MEK is downstream of BRAF, the MEK inhibitor trametinib (Mekinist) was developed as a way of blocking signalling in BRAF-V600 tumours pmc.ncbi.nlm.nih.gov.

Evidence behind trametinib in BRAF-V600E melanoma

- **Single-agent activity.** In the phase-III METRIC trial, patients with unresectable or metastatic melanoma carrying a BRAF V600E/K mutation were randomized to trametinib or chemotherapy. Median progression-free survival (PFS) was **4.8 months** with trametinib versus **1.5 months** with dacarbazine or paclitaxel pmc.ncbi.nlm.nih.gov. Six-month overall survival was **81 %** with trametinib and **67 %** with chemotherapy (even with cross-over) pmc.ncbi.nlm.nih.gov. On the basis of this improvement in PFS and overall survival, the U.S. FDA approved trametinib for unresectable or metastatic melanoma with a **BRAF V600E or V600K mutation** pmc.ncbi.nlm.nih.gov.
- **Combination with the BRAF inhibitor dabrafenib.** MEK inhibition alone produces only modest responses, so combination therapy is now the standard of care. In an early phase-I/II trial, the **dabrafenib + trametinib** arm had a median PFS of **9.4 months** compared with **5.8 months** for dabrafenib monotherapy and a response rate of **76 %** versus **54 %** pmc.ncbi.nlm.nih.gov. The later phase-III COMBI-d study compared dabrafenib plus trametinib with dabrafenib monotherapy in previously untreated BRAF V600E/K metastatic melanoma. At three years' follow-up, PFS was **22 %** versus **12 %**, and overall survival was **44 %** versus **32 %** for the combination versus monotherapy pmc.ncbi.nlm.nih.gov. In a favourable subgroup (normal lactate-dehydrogenase and <3 metastatic organs), three-year overall survival reached **62 %** with combination therapy pmc.ncbi.nlm.nih.gov. The combination also reduces skin toxicities and delays the emergence of resistance pmc.ncbi.nlm.nih.gov.
- **Safety considerations.** Common adverse effects of trametinib include rash, diarrhea, peripheral oedema and fatigue; when combined with dabrafenib, pyrexia is prominent pmc.ncbi.nlm.nih.gov. However, cutaneous squamous-cell carcinoma is less frequent in the combination arm than with a BRAF inhibitor alone pmc.ncbi.nlm.nih.gov.

CIViC interpretation

CIViC's clinical-significance field for *BRAF V600E – Melanoma – Trametinib* is built from these trial results. The mutation is considered **predictive for sensitivity** to MEK inhibition; evidence from randomized trials shows that trametinib, especially when combined with a BRAF inhibitor, improves progression-free and overall survival in patients whose melanoma harbours the BRAF V600E mutation pmc.ncbi.nlm.nih.gov pmc.ncbi.nlm.nih.gov. Consequently, CIViC classifies *BRAF V600E* in melanoma treated with trametinib as having **strong clinical significance** (evidence level A/1); the variant is an FDA-approved biomarker for treatment selection (dabrafenib + trametinib) and predicts patient benefit pmc.ncbi.nlm.nih.gov pmc.ncbi.nlm.nih.gov.

Sources

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